

NAME: _____

PERIOD: _____

One Line, Two Model Diagnostics

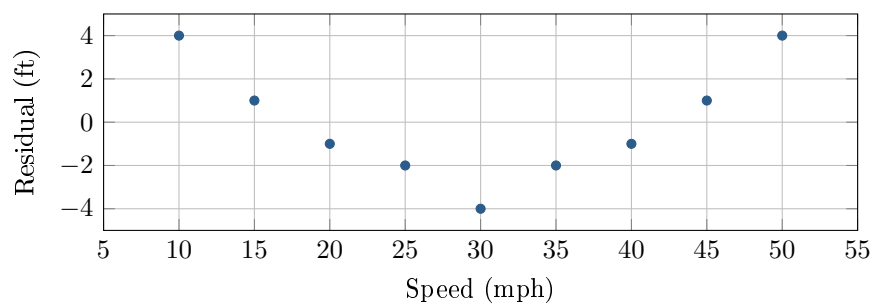
AP Statistics · Unit 5 · Topic 5.4

[STAR] THE PROBLEM

Suppose a road-safety lab records braking distance y at speeds x from 10 through 50 mph on one vehicle and dry surface. The line is $\hat{y} = -10 + 2.5x$, the correlation is $r = 0.997$, and the residual plot is shown. A residual is actual distance minus the line's prediction. At 50 mph, the actual distance is 119 feet.

Ezra: The large correlation and residuals on both sides of 0 justify using the line across the tested speeds.

June: I would use where the residuals occur and the 50-mph error to judge safety, then check any replacement model separately.



FIRST TAKE — one sentence before you work the parts

Would you trust this line to plan a safe stopping distance? Give one number or visible pattern in one sentence.

[BOOK] READING A RESIDUAL PLOT (keep on the page)

A residual is $y - \hat{y}$. A positive residual means the model underpredicted; a negative residual means it overpredicted. Plot residuals against x or $\hat{y}$. Random scatter around 0 supports a linear form. Curvature, trends, clusters, or changing spread show structure the line missed. Correlation measures the direction and strength of linear association, but even a large $|r|$ does not replace checking residual locations. Check a replacement model with its own residual plot.

Work (a)–(c).

DOK 1 (a) Read the residual at 50 mph. Does the line predict too high or too low there?

DOK 2 (b) Verify the 50-mph residual using the equation. Describe where the residual plot is above and below zero.

DOK 3 (c) Whose judgment is supported? Explain why the residual pattern matters even with the large correlation, and connect the 50-mph error to safety. State what you would check in a replacement model for this vehicle and dry surface. Write 3–4 sentences.

Frame: The correlation summarizes _____; the residual locations show _____.

Frame: At 50 mph the line predicts _____ while the actual value is _____, so for safety I would _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.